Fake News Detection Case Study Rubric

DS 4002 – Spring 2026 – Lauren Medica

Due: TBD

Submission format: Upload link to GitHub repository on Canvas

Individual Assignment

Why am I doing this?

This case study allows you to leverage your data science knowledge by applying natural language processing and supervised machine learning to a real-world problem: detecting fake news. As you work through this assignment, you will be exposed to how text data can be transformed into meaningful features and how machine learning models can be compared to answer a practical question about information and society.

What am I going to do?

The GitHub repository for this case study can be found at https://github.com/laurenmedica/Case-Study/tree/main. You will use a labeled dataset of over 40,000 news articles classified as fake or real. Using Python, you will preprocess both the article titles and full text, vectorize them using TF-IDF, and train two Random Forest classifiers, one on full article text and one on titles only. You will then evaluate both models using the F-1 score and compare their performance to determine how much information a headline alone contains.

Your final deliverables should include:

- Well documented, commented source code in a Jupyter Notebook
- At least three visualizations (e.g., word frequency charts, confusion matrices, model comparison)
- A REFERENCES.md file
- A GitHub repository containing all materials used

Tips for success:

- Apply the same preprocessing pipeline and train/test split to both models so the comparison is valid
- Use a fixed random seed (random_state=42) throughout so your results are reproducible
- Comment your code clearly, explain not just what you are doing but why

Make sure your visualizations are labeled with titles, axis labels, and legends **How will I know I have succeeded?**

You will meet expectations on this case study when you successfully follow and complete the criteria in the rubric below:

Spec Category	**Spec Details**
Formatting	● One GitHub repository (submitted via link on Canvas) ○ Create a new GitHub repository for this assignment titled 'CS3_FakeNews'

	● README.md ● LICENSE.md ● Source code file (Jupyter Notebook) ● DATA folder with your dataset ● REFERENCES.md
README.md	Brief summary of what you have produced for this case study. Should provide enough information to orient someone to your repository and explain how to run the code.
Source Code File	Well-documented Jupyter Notebook that contains all code used for the analysis. Must include: ● Exploratory data analysis: distribution of labels, article length comparisons, most common words by class, subject breakdown ● Text preprocessing applied consistently to both title and full text fields (lowercase, remove punctuation, tokenize, remove stopwords, stem) ● TF-IDF vectorization for both fields ● 80/20 train/test split with random_state=42 ● Random Forest classifier trained on full article text; evaluated with F-1 score (target: ≥ 0.90) ● Random Forest classifier trained on titles only; evaluated with the same metrics ● At least three visualizations (word frequency charts, confusion matrices, model comparison bar chart) ● Comments throughout explaining your process and interpreting results
REFERENCES.md	A Markdown file titled 'REFERENCES.md' citing any resources referenced in completing this case study in IEEE citation style. Include a brief annotation under each citation explaining how it informed your work.

Acknowledgments: Thank you to Professor Siller for providing the rubric structure!